Port Everglades Currents

Wind model, ship-type windage presets, and factors to improve — research brief for the current-buoy app

Prepared 1 November 2026 for Capt. Ryan Sweeney. Everything here describes a prototype for situational awareness. Nothing in it is a port rule, a pilot-group standard, or a certified navigation product.

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1. What the app does with the buoy data

The app reads NOAA PORTS for Port Everglades every six minutes: the Sea Buoy (LB PE, station PE0101, 16 one-metre depth bins from about 11 ft to 60 ft) and Buoy 2 (LB 2, station PE0201, 10 bins to about 41 ft), the South Port Everglades meteorological and water-level station (8722956), the Turning Basin tide predictions (8722951), and the NOAA current predictions inside the port (PEV0901 Entrance, PEV0902 Turning Basin, PEV0903 17th Street Bridge, PEV0905 South Port). See NOAA PORTS Port Everglades.

For each buoy the current vector (set, drift) is split onto the inbound course (270° T by default): the **cross** component sets the ship sideways and the **along** component adds to or subtracts from speed over ground. The headline value is the 14 ft "RT" bin NOAA publishes; the heading to steer uses the vector average of all bins within the entered draft, because the hull answers to the water it actually sits in, not just the surface layer. The crab angle is $\sin^{-1}$(cross ÷ speed through the water). Status (Normal / Caution / Over limit) uses the larger of the 14 ft cross and the draft-averaged cross; the thresholds are editable placeholders, not rules.

The Port Everglades PAWSA report describes why this matters here: a cross current of 2 to 2.5 kn between the entrance and the sea buoy, roughly a quarter of approach speed, that can set north or south depending on the Gulf Stream, can change direction up to three times during a transit, and can reach 5 kn at and beyond the sea buoy, with counter-currents at depth so that "the ship hull responds to the stronger current." (USCG PAWSA Port Everglades)

2. The wind model behind the ship-type selector

The original Low / Med / High windage switch has been replaced by a ship-type selector, with Low / Med / High kept as quick classes mapped to representative ships. The selected ship supplies a lateral wind area A_L, a length, beam and block coefficient, and a Blendermann aerodynamic class. The estimate then runs in four steps:

- **Apparent wind.** True wind from South Port (speed, direction, gust) is combined vectorially with the ship's own speed along the course. Apparent wind speed W_a and its angle γ off the bow drive the side force; a following wind on an inbound ship is partly cancelled by ship speed, a head wind is amplified.
- **Side force on the ship.** $Y_{wind} = \tfrac{1}{2}\,\rho_a\, C_Y(\gamma)\, A_L\, W_a^2$, $\rho_a = 1.25$ kg/m³ as in PIANC 2014 (PIANC Report 121). $C_Y(\gamma) = C_{Dt} \sin\gamma \div [1 - \tfrac{1}{2}\,\delta\,(1 - C_{Dl}/C_{Dt})\sin^2 2\gamma]$ with Blendermann's 1994 parameters by ship type, e.g. C_{Dt} 0.90 and δ 0.80 for a passenger liner, 0.95 / 0.80 for a car carrier, 0.90 / 0.40 for a loaded container ship, 0.70 / 0.40 for a tanker (Blendermann 1994 coefficients, MSS blendermann94.m).
- **Hull reaction at a drift angle.** Underway, the hull develops a sway force $Y_{hull} = \tfrac{1}{2}\,\rho_w\, L_{pp}\, T\, V^2\,[K\cdot Y'_v\cdot\beta + K\cdot C_D\cdot\beta^2]$. The linear term uses Clarke's 1983 regression $Y'_v = \pi\,(T/L)(1 + 0.4\,C_b\,B/T)$; the quadratic term is a crossflow drag $C_D \approx 0.8$ as used in the IALA IWRAP drift model (IALA IWRAP drift). K is Clarke's deep-to-shallow multiplier $K_0 + K_1(B/T) + K_2(B/T)^2$ with $F = h/T - 1$, $K_0 = 1 + 0.0775/F^2 - 0.0110/F^3 + 0.000068/F^5$, $K_1 = -0.0643/F + 0.0724/F^2 - 0.0113/F^3 + 0.0000767/F^5$, $K_2 = 0.0342/F$, valid for h/T above about 1.25 and clamped there (ITTC Manoeuvring Committee summary of Clarke 1983; Lee et al. 2009). Applying K to the quadratic term as well is a simplification and is flagged as an assumption in the code.
- **Solve for leeway.** Setting $Y_{hull} = Y_{wind}$ gives a quadratic in β; its root is the leeway angle. The heading correction is $-\beta$ (steer into the wind), added to the current crab. Steady wind and gust are both solved so the app shows a range, e.g. "With wind 272–273°". When ship speed is under about half a knot the steady solution is meaningless, so the app shows instead the stopped-ship drift from Richardson's balance v = W

$\sqrt{[(\rho_a/\rho_w)(C_{da}A_a)/(C_{dw}A_w)]}$ (Richardson 1997).

Sanity checks with live data (1 Nov 2026, wind 21–25 kn from ENE/NE, inbound at 10–12 kn, 28 ft draft):

Ship preset	A_L (m²)	Leeway, steady–gust	Stopped drift	Reads as
Cruise, Oasis / Icon class, Sea Buoy (118 ft)	15,750	1.0–1.4° to port (21 kn quartering)	1.6 kn	With 0.6° current crab: steer 268°
Cruise, mid-size, Sea Buoy	11,950	2.8–3.5° (25 kn, 12 kn ship)	1.6 kn	Steer 272–273° vs 269° current only
Cruise, Oasis class, Buoy 2 (55 ft, h/T 2.0, K = 2.14)	15,750	2.8–3.5° (25 kn, 10 kn ship)	1.5 kn	Shallow water roughly halves the angle a deep-water balance would give
Car carrier PCTC 6,000 CEU, loaded	4,850	1.1–1.8°	1.3 kn	Steer 270–271°
Container Panamax 4,000 TEU	4,500	0.3–0.5° (21 kn)	1.1 kn	Rounds to the current-only heading
Product tanker MR, loaded	1,900	0.1°	0.7 kn	Negligible, as the Low class implies
Beam wind 25 kn, Oasis class, 10 kn, 55 ft water	15,750	4.5°	—	Order of magnitude pilots report for large cruise ships

These are model outputs, not measurements. The Settings screen has a **wind gain** multiplier so a pilot who consistently sees more or less leeway than the app predicts can scale the South Port wind without touching the model.

3. Ship-type windage presets and their sources

Lateral (broadside) wind areas are taken from published design tables rather than estimated. Frontal areas enter only the small Blendermann cross-coupling term and are approximated from the frontal-to-lateral ratio of the type (0.21 for Oasis of the Seas, which Thoresen lists at 15,764 m² lateral and 3,354 m² frontal). Where a preset interpolates between table rows or is an estimate, the app says so.

Preset	L_{pp} × B (m)	A_L loaded / ballast (m²)	Class	Source row
Cruise · Oasis / Icon class (~225k GT)	333 × 47	15,750 / —	High	Thoresen measured 15,764; Trelleborg 220k GT row 15,700
Cruise · large (~140–160k GT)	313.6 × 43.7	13,800 / —	High	Trelleborg cruise (post-Panamax) 160k GT
Cruise · mid-size (~90–115k GT)	290 × 36	11,950 / —	High	Trelleborg cruise 115k GT
Cruise · small (~50k GT)	212 × 28	6,850 / —	High	Trelleborg cruise (Panamax) 50k GT
Car carrier · PCTC ~6,000 CEU	189 × 32.2	4,850 / 5,800	High	Trelleborg car carrier 57k DWT
Car carrier · PCTC ~8,000 CEU	210 × 32.2	5,700 / 6,900	High	Trelleborg car carrier 70k DWT
Ro-ro freight / ferry (~30k DWT)	237 × 36.4	4,300 / 4,500	High	Trelleborg conventional ferry 30k DWT
Container · post-Panamax ~8,000 TEU	323 × 43.2	7,200 / 7,800	Med	Trelleborg container 110k DWT
Container · Panamax ~4,000 TEU	253 × 32.2	4,500 / 4,700	Med	Trelleborg container 50k DWT
Container · feeder ~1,000 TEU	145 × 23.4	3,100 / 3,200	Med	Trelleborg 10k DWT / 1,000 TEU
General cargo · ~20k DWT	158 × 24.8	2,100 / 2,800	Med	Trelleborg general cargo 20k DWT
Product tanker · MR ~50k DWT	174 × 32.2	1,900 / 3,000	Low (Med in ballast)	Trelleborg product tanker 50k DWT

Preset	L_{pp} × B (m)	A_L loaded / ballast (m^2)	Class	Source row
Tanker · LR1 / Panamax ~75k DWT	219 × 32.2	2,350 / 3,850	Low (Med in ballast)	Interpolated, Trelleborg 60k and 80k DWT
Tanker · Aframax ~100k DWT	239 × 44	2,750 / 4,500	Low (Med in ballast)	Trelleborg tanker 100k DWT
Bulk carrier · Panamax ~60k DWT	210 × 33.5	2,050 / 3,500	Low (Med in ballast)	Trelleborg bulk carrier 60k DWT
Tug & petroleum barge (ATB ~185 m)	175 × 24	900 / 1,700 (estimate)	Low	No published row; barge length × freeboard plus tug house
Custom	user	user	Med	Entered in Settings

Sources: Trelleborg Fender Application Design Manual (ship tables); Thoresen, Port Designer's Handbook (Oasis class, container and tanker areas). Cross-checks: PIANC 2014 Table 2-3 regressions $A = a \cdot X^b$ from Akakura & Takahashi 1998 (PIANC Report 121) and the NILIM 50 % / 75 % tables (NILIM Technical Note 309). The Trelleborg tanker and bulker "ballast" rows are roughly 1.6 to 1.7 times the loaded area, which is why a tanker in ballast moves from Low to Med in the app.

4. Other factors worth adding

Ranked by how much they would change a heading or a decision at Port Everglades, with the data source each would need.

1. **Counter-currents at depth and shear across the draft.** The PAWSA report notes opposite-sign currents at depth; the app already averages bins within the draft, but it should flag when the surface and deep bins disagree by more than a set amount (for example 0.5 kn or 60°) because the headline 14 ft value is then misleading. Source: existing PE0101 / PE0201 bins.

2. **Along-channel current into speed over ground and time to the jetties.** A 1 kn following current on a 10 kn ship is 10 % more ground speed and less rudder effect per yard; USACE requires at least 4 kn of speed through the water above the current for steerage (USACE EM 1110-2-1613). Show SOG and minutes from each buoy to the jetty tips.

3. **Wind yaw moment and pivot point.** The current model balances side force only. Wind on a cruise ship's superstructure also yaws the hull; whether the bow pays off or into the wind depends on the centre of wind pressure relative to the pivot point, and it changes with speed. Blendermann's 1994 set includes a yaw coefficient (κ) that could be added, with the ship's A_L centroid position as one more preset field.

4. **Gusts and averaging interval.** South Port reports a 6-minute mean and a gust. The app solves both; a next step is to weight by how long a gust lasts versus the ship's response time (tens of seconds for a large ship), so the gust heading is shown as a bound rather than a target.

5. **Wind sensor height, exposure and the 1/7-power correction.** ITTC recommends scaling wind to the ship's reference height with $U(z) = U(10)\,(z/10)^{1/7}$ (ITTC 7.5-04-01-01.2). The South Port anemometer sits inside the harbour; the sea buoy is open water. The wind-gain setting is the stopgap; a licensed WeatherFlow or an entrance anemometer would be the fix.

6. **Waves and swell.** Onshore wind against the ebb produces 4 to 6 ft seas in the entrance (PAWSA). NDBC 41122 off Hollywood Beach is not readable directly from the browser; a small Netlify function could proxy it and add wave height, period and direction to the Now screen (NDBC 41122).

7. **Shallow-water and bank effects in the jetties.** PAWSA describes inner-south and outer-north cross currents at the jetties on the flood and hard rock banks; squat and bank suction belong to the UKC app, but a

one-line link from each buoy to the UKC calculation (same draft, same tide) would keep the two tools consistent.

8. **Predicted versus observed inside the port.** In-port currents are NOAA predictions; the entrance prediction (PEV0901) can be compared with Buoy 2 each refresh to show how much today's observed current departs from the table.

9. **Outbound mode.** Added: a transit toggle that uses the reciprocal course and relabels inbound/outbound, so an outbound pilot gets the same crab and leeway logic.

10. **Calibration log.** A pilot who records the heading actually held against the app's prediction for a given ship, wind and speed will quickly show whether the wind gain should be above or below 1.0 by ship class. Storing a few dozen such pairs on the phone would turn the model from literature values into Port Everglades values.

11. **Design-limit context.** PIANC 2014 uses 15 m/s (29 kn) wind and 1 m/s (2 kn) cross current at half draught as typical channel design limits and allows 5° to 20° drift angle depending on depth-to-draught; those are useful references for the Caution / Limit settings, not rules for this port.

5. Limits of the estimate

The leeway model is a steady-state balance with literature coefficients. It ignores wind yaw moment, rudder-induced drift, waves, hull fouling and the actual superstructure of the ship alongside; the lateral areas are class averages from design tables. Clarke's shallow-water factors are regressions valid for h/T above about 1.25, and applying the same factor to the crossflow term is a simplification. The South Port wind is sheltered relative to the sea buoy. For these reasons the app rounds to whole degrees, shows steady-to-gust ranges, labels the ATB preset as an estimate and keeps the pilot's judgment as the authority.

6. Sources

NOAA CO-OPS PORTS, Port Everglades: https://tidesandcurrents.noaa.gov/ports/index.html?port=pe

USCG PAWSA workshop report, Port Everglades: https://navcen.uscg.gov/sites/default/files/pdf/pawsa/workshopReports/Port%20Everglades.pdf

PIANC Report 121, Harbour Approach Channels Design Guidelines (2014): https://shaghool.ir/Files/NAVIGATION-PIANC-Harbour-Approach-Channels-Design-Guidelines-2014.pdf

Blendermann (1994) wind-load coefficients as tabulated in Fossen's MSS toolbox: https://raw.githubusercontent.com/cybergalactic/MSS/master/LIBRARY/environment/blendermann94.m

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IALA IWRAP, contact and drift model: https://www.iala.int/wiki/iwrap/images/6/65/Contact_drift.pdf

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NDBC station 41122, Hollywood Beach: https://www.ndbc.noaa.gov/station_page.php?station=41122